Solar Panel Thermal Anomaly Detection System

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Project Type: Deep Learning / Computer Vision / Renewable Energy AI

1. Executive Summary

Photovoltaic (PV) solar panels are susceptible to various physical and electrical anomalies (such as hot spots, cell cracks, and diode failures) that reduce energy efficiency and can cause permanent damage. This project implements an automated computer vision pipeline using infrared (IR) thermal imagery and **Transfer Learning (MobileNetV2)** to classify solar modules into a clean binary framework: **Normal (No-Anomaly)** vs **Defective/Anomalous (between 11 different types of anomalies)**

2. Dataset & Preprocessing

- **Source:** Open-source *Raptor Maps Infrared Solar Modules* dataset.
- **Metadata Processing:** Automatically clones the official repository, parses `module_metadata.json`, and transposes the structure so each row corresponds to a unique solar panel module.
- **Class Mapping:** Aggregates 12 distinct multi-class anomaly categories (Hot-Spot, Cracking, Soiling, Cell-Multi, etc.) into a robust binary classification task:
  - `0`: `No-Anomaly` (Healthy panels)
  - `1`: Any specific anomaly type
- **Stratified Splitting:** To avoid data leakage and maintain class proportions, a strict 3-way split is performed using Scikit-Learn:
  - **Training Set (72%)**: Used for model weight optimization.
  - **Validation Set (18%)**: Used for hyperparameter tuning and early monitoring.
  - **Test Set (10%)**: Completely isolated blind test set to evaluate real-world generalization.

3. Model Architecture (Transfer Learning)

The model leverages a pre-trained **MobileNetV2** backbone optimized for lightweight, efficient edge-device or server-side execution:

- **Base Model:** MobileNetV2 (ImageNet weights, frozen to retain generic feature extraction).
- **Custom Classification Head:**
  - `GlobalAveragePooling2D`: Reduces spatial dimensions of feature maps.
  - `Dense(64, activation='relu')`: Fully connected layer for pattern synthesis.
  - `Dropout(0.5)`: Regularization technique to prevent overfitting.
  - `Dense(1, activation='sigmoid')`: Output probability layer for binary classification.
- **Compilation Optimizer:** Adam optimizer with `binary_crossentropy` loss and accuracy metrics.

4. Installation & Usage

Prerequisites

Ensure Python and pip are installed, then install the required libraries:

```
pip install tensorflow pandas scikit-learn matplotlib
```

Execution

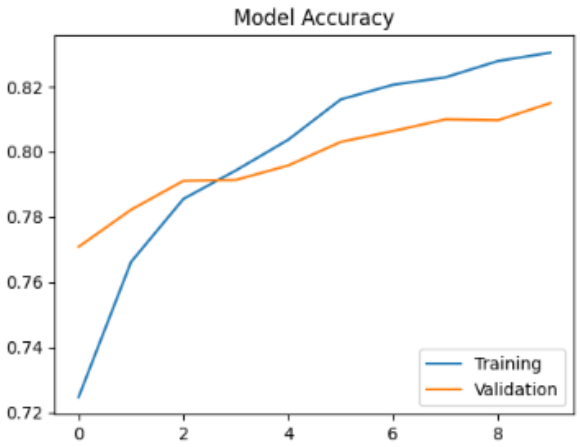
Run the main script to execute data acquisition, preprocessing, training, evaluation, and visualization:

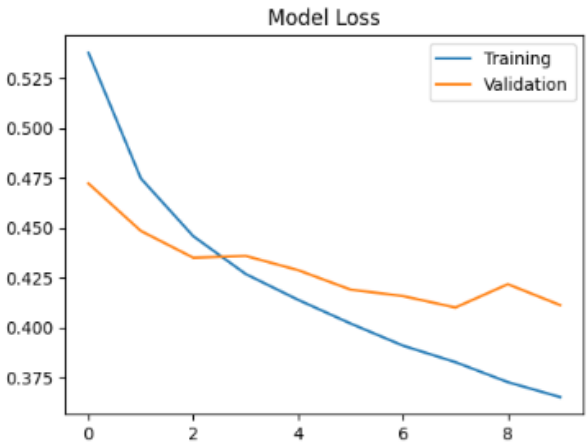
```
python Inspector_solar.py
```

5. Results & Visual Diagnostics

The pipeline automatically generates:

1. **Training Curves:** Side-by-side plots tracking Model Accuracy and Loss across epochs.





2. **Blind Test Evaluation:** Quantitative metrics (Loss & Accuracy) on the isolated test set.

3. **Visual Prediction Grid:** A multi-panel graphical display showing real thermal images alongside ground-truth labels, predicted labels, and confidence probabilities (color-coded in green for correct predictions and red for errors).

